

AMATH 353 HW 7

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Question 1: An Easier Shock Solution

One setting where the method of characteristics fails is when the characteristics intersect and the solution becomes infinitely steep. This leads to the formation of *shock wave*, and we saw that the shock moves along a characteristic $(x_s(t), t)$ which can be determined from Rankine-Hugoniot condition

$$\frac{dx_s(t)}{dt} = \frac{\phi(u^+) - \phi(u^-)}{u^+ - u^-}.$$

Enforcing the Rankine-Hugoniot condition, determine the shock solution of the IVP

$$\begin{cases} u_t + uu_x = 0, & x \in \mathbb{R}, t \geq 0, \\ u(x, 0) = u_0(x), & x \in \mathbb{R}, \end{cases} \quad \text{where} \quad u_0(x) = \begin{cases} 2, & x < -1, \\ 1 - x, & |x| \leq 1. \\ 0, & x > 1. \end{cases}$$

Find the characteristic curve.

$$\begin{aligned} u_t + uu_x &= 0 \\ c(u) &= u \\ \frac{dx}{dt} &= u \\ u &= u_0(x_0) \\ \frac{dx}{dt} &= u \\ dx &= u dt \\ \int dx &= \int u dt \\ x(t) &= u_0(x_0)t + C \end{aligned}$$

Solving for C

$$\begin{aligned} x(0) &= x_0 \\ x_0 &= u_0(x_0)(0) + C \\ x_0 &= C \end{aligned}$$

$$x(t) = u_0(x_0)t + x_0$$

Question 2: A Harder Shock Solution

Enforcing the Rankine-Hugoniot condition, determine the shock solution of the IVP

$$\begin{cases} u_t + uu_x = 0, & x \in \mathbb{R}, t \geq 0, \\ u(x, 0) = u_0(x), & x \in \mathbb{R}, \end{cases} \quad \text{where} \quad u_0(x) = \begin{cases} 0, & x < 0, \\ x, & 0 \leq x \leq 1. \\ 0, & 1 < x. \end{cases}$$

Question 3: Three Shocks

Enforcing the Rankine-Hugoniot condition, determine the shock solution of the IVP

$$\begin{cases} u_t + u^2 u_x = 0, & x \in \mathbb{R}, t \geq 0, \\ u(x, 0) = u_0(x), & x \in \mathbb{R}, \end{cases} \quad \text{where} \quad u_0(x) = \begin{cases} 2, & x < 0, \\ 1, & 0 \leq x \leq 1. \\ 0, & 1 < x. \end{cases}$$

Question 4: Two shocks Enforcing the Rankine-Hugoniot condition, determine the shock solution of the IVP

$$\begin{cases} u_t + uu_x = 0, & x \in \mathbb{R}, t \geq 0, \\ u(x, 0) = u_0(x), & x \in \mathbb{R}, \end{cases} \quad \text{where} \quad u_0(x) = \begin{cases} 1, & x < -1, \\ 0, & -1 \leq x \leq 1. \\ -1, & 1 < x. \end{cases}$$

Question 5: Feedback

How long did you spend on this homework? Was it easy/medium/hard? Do you have any feedback for the instructor/TA? Do you like the inclusion of bonus problems?

This homework took me a little while but that was because I was busy with work from other classes. I liked how the problems on this homework were mostly the same style. I have no opinion on the bonus questions since I normally don't have time to do them.